THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 1 & 2: **Real Numbers, Polynomials** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The prime factorisation of 7429 is: 1

(a) 17×437 **(b)** $7 \times 11 \times 97$ **(c)** $13 \times 19 \times 29$ **(d)** $17 \times 19 \times 23$

2. The LCM of 17 and 19 is: 1

(a) 1 **(b)** 323 **(c)** 17 **(d)** 36

3. The product of the HCF and LCM of 12 and 18 is: 1

(a) 36 **(b)** 216 **(c)** 108 **(d)** 6

4. A student states: "A composite number can be factorised into primes in more than one way (ignoring order)." The correct response is: 1

(a) Correct for even numbers
(b) Correct
(c) Incorrect — the factorisation is unique (FTA)
(d) Cannot be determined

5. For $\sqrt{12}$, Student A says $2\sqrt{3}$ (irrational) and Student B says 6 (rational). Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. A quadratic polynomial whose zeroes are 0 and 4 is: 1

(a) $x^2 + 4x$ **(b)** $x^2 - 4$ **(c)** $x^2 - 4x$ **(d)** $x^2 + 4$

7. If α, β are the zeroes of $x^2 - 2x - 8$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 16 **(b)** 10 **(c)** −6 **(d)** −16

8. The zeroes of $x^3 - 4x$ are: 1

(a) $0, 4$ **(b)** $0, 2, -2$ **(c)** $2, -2$ **(d)** $4, -4$

9. If the zeroes of $ax^2 + bx + c$ are reciprocals of each other, then: 1

(a) $b = c$ **(b)** $a = c$ **(c)** $a = b$ **(d)** $a = -c$

10. A student states: "A linear polynomial can have two zeroes." The correct response is: 1

(a) Incorrect — it has exactly one zero **(b)** Correct if the constant is 0
(c) Correct **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the HCF and LCM of 8, 9 and 25. 2

12. Given that LCM$(306, 657) = 22338$, find HCF$(306, 657)$. 2

13. Find a quadratic polynomial whose sum and product of zeroes are 0 and $\sqrt{5}$ respectively. 2

14. Check whether -2 and 3 are zeroes of $x^2 - x - 6$, and verify the sum of zeroes. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $2 + 5\sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational. 3

16. Find the zeroes of $v^2 + 4\sqrt{3}v - 15$ and verify the relationship between the zeroes and the coefficients. 3

17. The length, breadth and height of a room are 8.25 m, 6.75 m and 4.50 m. Find the length of the longest rod that can measure all three exactly. 3

OR

Find the HCF and LCM of 96 and 360 using prime factorisation. 3

18. If one zero of $(a^2 + 9)x^2 + 13x + 6a$ is the reciprocal of the other, find a. 3

OR

If α and β ($\alpha > \beta$) are the zeroes of $x^2 + 7x + 10$, find $\alpha - \beta$ without finding the zeroes. Verify by finding them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A librarian has 336 Mathematics books and 240 Science books. She wants to stack them so that each stack has the same number of books of one subject only, using the least number of stacks. 5

(i) Find the number of books in each stack.
(ii) Find the total number of stacks.
(iii) If each book is 2 cm thick, find the height of each stack.

OR

(i) Find the HCF and LCM of 72 and 120. 5
(ii) Find $\frac{\text{LCM}}{\text{HCF}}$.
(iii) Can two numbers have HCF 24 and LCM 540? Justify.

20. **Case study:** The profit (in thousand ₹) of a firm making x hundred units is $P(x) = -x^2 + 10x - 21$. 5

(i) Find the break-even points (zeroes).
(ii) Find the profit at $x = 5$.
(iii) Verify the relationship between the zeroes and coefficients.

OR

Two zeroes of $x^3 - 4x^2 - 3x + 12$ are $\sqrt{3}$ and $-\sqrt{3}$. Find the third zero and verify the relationship $\Sigma\alpha\beta = \frac{c}{a}$. 5

END OF QUESTION PAPER — Total: 40 Marks